

AI Engineering Intensive

From LLM Foundations to Production-Grade AI Systems

COMPLETE SYLLABUS · WEEKS 1–12 · 84 DAYS

6 July 2026 Series start	8 November 2026 Series end	84 days · 12 weeks 18 calendar weeks
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Structure

Phase	Weeks	Days	Dates	Status
Phase 1	1–4	1–28	6 Jul – 2 Aug 2026	■ COMPLETE
■■ Pause	—	—	3 Aug – 13 Sep 2026 (42 days)	In progress
Phase 2	5–12	29–84	14 Sep – 8 Nov 2026	Upcoming

Topics. LLMs · Tokenization · Embeddings · Transformers · Quantization · Fine-Tuning · Serving · RAG · Hybrid Retrieval · Agents · Tool Use · Multi-Agent · LLMOps · Versioning · Cost Engineering · Security · Multimodal · **Harness Engineering** · **Loop Engineering** · **Context Engineering** · MCP · Evals · Guardrails · System Design · Red Teaming · Cloud Deployment

Phase 1 verified: 28 lesson PDFs · 549 pages · 371 Eagle Eye terms · 0 gaps · 0 FAANG-rule violations. Running capstone (Days 13–27): a production Text-to-SQL system at **82.8% execution accuracy, p95 3,740 ms, \$0.002694/query**.

PHASE 1 — COMPLETE ■

Weeks 1–4 · Days 1–28 · 6 July – 2 August 2026

WEEK 1 · FOUNDATIONS

Days 1–7 · 6–12 July 2026 · Project: Model Selection Dashboard (MINI)

Day	Date	Topic	The one thing that matters
1	Mon Jul 6	Tokenization	Tokens are the billing unit <i>and</i> the context unit
2	Tue Jul 7	Embeddings	Similarity is geometry; cosine over normalized vectors
3	Wed Jul 8	Transformer Architecture	Attention is quadratic — why long context costs
4	Thu Jul 9	Training Lifecycle	Pre-train vs fine-tune vs prompt: different problems
5	Fri Jul 10	Scaling Laws & Prompt Eng.	CoT is the cheapest accuracy in the curriculum
6	Sat Jul 11	Structured Output	Constrain the format, don't ask for it
7	Sun Jul 12	Model Selection Dashboard	Weighted criteria, not "which is best"

Also covered: BPE/WordPiece, vocabularies, attention masks · cosine similarity, dense retrieval, embedding space · self-attention, multi-head, positional encoding, RoPE, feed-forward, layer norm, residuals · backpropagation, optimizers, LR schedules, checkpointing · Chinchilla, compute-optimal, emergent abilities, few-shot/zero-shot, in-context learning · Pydantic, JSON schema, constrained decoding, function calling · benchmark/latency/cost/control matrix

WEEK 2 · EFFICIENCY

Days 8–14 · 13–19 July 2026 · Project: Domain Fine-Tune + Quantize (PORTFOLIO)

Day	Date	Topic	The one thing that matters
8	Mon Jul 13	Quantization	4×–8× memory reduction, minimal quality loss
9	Tue Jul 14	LoRA & PEFT	Adapters make specialization economically viable
10	Wed Jul 15	Full Fine-Tuning	Catastrophic forgetting is real and measurable
11	Thu Jul 16	Model Serving	Continuous batching: 210 → 900 q/hr
12	Fri Jul 17	Evaluation	Pick the metric that matches the failure mode
13	Sat Jul 18	Efficiency	PagedAttention enables prefix sharing
14	Sun Jul 19	Text-to-SQL Foundation	Execution accuracy, not exact match

■ **The running capstone begins Day 13** — a production Text-to-SQL system built continuously through Day 27, carrying measured numbers forward day by day.

Also covered: INT8/INT4, GGUF, AWQ, GPTQ, weight compression · rank, alpha, target modules, QLoRA, adapter merging · warmup, gradient accumulation, mixed precision, DeepSpeed, FSDP · vLLM, Triton, FastAPI, KV cache · BLEU/ROUGE, perplexity, Elo/arena, RAGAS · speculative decoding, FlashAttention, PagedAttention, tensor/pipeline parallelism · SPIDER benchmark, NL2SQL, Prometheus, SLOs

WEEK 3 · RETRIEVAL & AGENTS

Days 15–21 · 20–26 July 2026 · Project: End-to-End Text-to-SQL (CAPSTONE)

Day	Date	Topic	The one thing that matters
15	Mon Jul 20	RAG Foundations	Retrieval adds knowledge; nothing else does
16	Tue Jul 21	RAG Indexing Pipelines	Fingerprints make re-indexing idempotent
17	Wed Jul 22	RAG Retrieval Deep Dive	Dense + BM25 + RRF + MMR + rerank
18	Thu Jul 23	RAG Generation & Prompting	CoT + faithfulness + self-correction + citations
19	Fri Jul 24	Agents and Tool Use	Agent = loop; MAX_STEPS is a safety control
20	Sat Jul 25	Multi-Agent Systems	Specialize by role → mix model sizes → cut cost
21	Sun Jul 26	Week 3 Capstone	Ablation justifies architecture: RAG = +53.5pp

Also covered: chunking, vector stores, Qdrant, context windows · idempotent upsert, multi-tenant namespaces · parent-child, multi-vector, query expansion, MMR, cross-encoder reranking · sqlglot faithfulness enforcement, dynamic few-shot, dialect calibration · ReAct, tool schemas, working + long-term memory · orchestrator-worker, parallel phases, agent pools, speculative execution · ablation study, cost model, production readiness checklist

WEEK 4 · PRODUCTION

Days 22–28 · 27 July – 2 August 2026 · Project: Production Hardening (PORTFOLIO)

Day	Date	Topic	The one thing that matters
22	Mon Jul 27	Production LLM Operations	Your failure mode is a 200 OK with a wrong answer
23	Tue Jul 28	Prompt & Model Versioning	A prompt is a model weight you can edit in a text editor
24	Wed Jul 29	Cost Optimization at Scale	73% of input tokens are identical every request
25	Thu Jul 30	Security and Compliance	Assume injection succeeds; design for a compromised model
26	Fri Jul 31	Multi-Modal Systems	An image is tokens; extract once, index forever
27	Sat Aug 1	Fine-Tuning for Production	Doesn't add knowledge; buys cost and latency
28	Sun Aug 2	Week 4 Capstone & Close	The handoff is the deliverable

Also covered: four failure classes, deployment topologies, canary suites, drift detection, incident triage · immutable artifact registry, golden-set CI gate, A/B testing, rollout patterns · prompt caching, semantic caching, complexity routing, cost attribution · prompt injection, PII handling, tenant isolation, residency, audit trails, supply chain · VLM architecture, document pipelines, table extraction · fine-tune decision framework, dataset construction, ship gate

Phase 1 Outcome — The Capstone System

Production Text-to-SQL, built continuously across Days 13-27:

82.8% execution accuracy (SLO floor 82%)

3,740 ms p95 latency (SLO 4,000 ms)
\$0.002694 per query (-57.2% from baseline)
1,240 queries/hour sustained

0.8pp accuracy budget remaining <- the binding constraint

Request path: input guardrails -> semantic cache -> artifact resolution ->
orchestrator -> parallel(retrieval, memory) -> complexity router ->
generation(8B or 70B) -> validation -> output guardrails -> execute

■ ■ PAUSE

3 August – 13 September 2026 · 42 days · no teaching days

On resume, before Day 29: read the Resume Brief (Day 28 §5) · re-read Days 19–20 (agents, multi-agent) — Week 5 builds directly on both · run one canary evaluation to confirm nothing drifted.

First three actions: (1) buy accuracy headroom — two prompt fixes worth ~3pp; (2) deploy Tier 1 security — `llm_query_role` + RLS, 20 min; (3) wire the two CI gates — golden set + `IsolationTestSuite`.

PHASE 2

Weeks 5–12 · Days 29–84 · 14 September – 8 November 2026

Content lock (2 Aug 2026). A coverage audit of the Phase 1 corpus found **Evals** and **Guardrails** strongly covered, but **Harness Engineering** a genuine gap, **Context Engineering** scattered, and **Loop Engineering** present as mechanics without being named as a discipline. Weeks 5 and 6 were rewritten day-by-day to teach all three explicitly — **no extra weeks added**.

WEEK 5 · AGENTS, HARNESS & LOOP ENGINEERING

Days 29–35 · 14–20 September 2026 · Project: Tool-Calling Agent (PORTFOLIO)

Day	Date	Topic	Focus
29	Mon Sep 14	Agent Architectures Beyond ReAct	Plan-execute, reflection, tree search, router-only; when each applies
30	Tue Sep 15	Harness Engineering I — Tool Design	Granularity, naming, namespacing, affordances, schema design, tool-selection accuracy
31	Wed Sep 16	Harness Engineering II — Responses & Errors	Response shape, truncation, error messages that teach recovery, sandbox as design surface
32	Thu Sep 17	Loop Engineering I — Termination & Budgets	Stopping conditions; steps/tokens/wall-clock/cost as a joint budget; convergence vs thrashing
33	Fri Sep 18	Loop Engineering II — Recovery & Sub-Agents	Retry taxonomy, escalation ladders, when to spawn a sub-agent, observation formatting
34	Sat Sep 19	Agent Frameworks & Human-in-the-Loop	Framework landscape, approval gates, interruption and resumption
35	Sun Sep 20	Week 5 Project — Tool-Calling Agent	Harness and loop <i>are</i> the deliverable, with measured budgets

WEEK 6 · CONTEXT ENGINEERING, MCP & MULTI-AGENT

Days 36–42 · 21–27 September 2026 · Project: MCP Server + Multi-Agent Pipeline (PORTFOLIO)

Day	Date	Topic	Focus
36	Mon Sep 21	Context Engineering I — The Context Budget	Window as budget to allocate; context rot; position effects; utilisation vs usefulness
37	Tue Sep 22	Context Engineering II — Compaction	Compaction, progressive disclosure, just-in-time retrieval, note-taking, sub-agent isolation
38	Wed Sep 23	MCP Fundamentals	Protocol, servers/clients, transports; resources vs tools vs prompts
39	Thu Sep 24	Building a Production MCP Server	Expose the capstone over MCP; auth, per-tenant scoping, contract versioning
40	Fri Sep 25	Multi-Agent Communication & Handoffs	Handoff protocols, shared vs isolated state, deadlock, trace debugging
41	Sat Sep 26	Agent Memory Architectures	Episodic/semantic/procedural, consolidation, cross-session, erasure by design
42	Sun Sep 27	Week 6 Project — MCP + Multi-Agent	Explicit context budget per agent; measured before/after utilisation

WEEKS 7–12 · Specified at Week Level

Each week is expanded day-by-day before it begins, so the expansion reflects what actually happened in the preceding weeks rather than guessing now.

WEEK 7 · EVALUATION & PRODUCTION APPLICATIONS

Days 43–49 · 28 Sep – 4 Oct 2026 · Project: Eval & Observability Platform (PORTFOLIO)

LLM-as-a-judge (the one eval technique absent from Phase 1), eval harness design, golden dataset construction, regression gates, tracing and observability tooling, user feedback loops, production dashboards.

Builds on: Days 12, 21, 22, 23

WEEK 8 · AGENTIC SYSTEM DESIGN

Days 50–56 · 5–11 Oct 2026 · Project: Scalable Agent Runtime (MINI)

Agent runtime architecture, state management, queueing and backpressure, horizontal scaling, failure isolation, concurrency at agent scale.

Builds on: Days 20, 22

WEEK 9 · IMAGE, MULTIMODAL & REASONING MODELS

Days 57–63 · 12–18 Oct 2026 · Project: Multimodal Document Processor (PORTFOLIO)

Vision-language architectures, document understanding at scale, OCR pipelines, chart and diagram reasoning, reasoning models and test-time compute.

Builds on: Day 26

WEEK 10 · AI SECURITY, RED TEAMING & RESPONSIBLE AI

Days 64–70 · 19–25 Oct 2026 · Project: Red Team Your Own RAG System (PORTFOLIO)

OWASP LLM Top 10, systematic adversarial testing (Garak, PyRIT), guardrail engineering, injection defences, bias and fairness metrics, model cards, and the regulatory landscape (EU AI Act, NIST AI RMF, ISO 42001). Deliverable: 20+ adversarial prompts against your own RAG service, guardrails blocking $\geq 90\%$, and a written security assessment with risk ratings and residual-risk acceptance.

Builds on: Day 25

WEEK 11 · CLOUD-NATIVE DEPLOYMENT, LLMOps & SYNTHETIC DATA

Days 71–77 · 26 Oct – 1 Nov 2026 · Project: Deploy a Production-Ready AI Service (PORTFOLIO)

Cloud deployment (AWS/GCP/Azure), GPU node pools with autoscaling, CI/CD for AI with automated eval gates, model registries, inference optimisation (TensorRT-LLM, ONNX), synthetic data generation (Self-Instruct, Evol-Instruct), feature stores and data lineage, cost engineering and chargeback.

Builds on: Days 11, 22, 23, 24, 27

WEEK 12 · CAPSTONE — ENTERPRISE AI ASSISTANT

Days 78–84 · 2–8 Nov 2026 · Project: Full Capstone (CAPSTONE)

One integrated system wiring RAG + Agents + Eval + Security + Cloud Deploy, defended as a full architecture proposal.

#	Required deliverable
1	Architecture Decision Records for every major choice (vector DB, model, framework, cloud)
2	C4 system diagrams — Context, Container, Component, Code
3	Non-functional requirements — latency SLAs, availability, cost/request, scaling limits
4	Risk assessment — failure modes, guardrail-bypass scenarios, residual risk
5	Deployment plan — blue-green strategy, rollback, monitoring dashboards
6	Security report — red-team results, guardrail coverage, compliance mapping

Discipline Coverage Across the Full Programme

Discipline	Phase 1	Phase 2	Dedicated days
Evals	D12, D21, D22, D23, D26, D27	Week 7 (7 days)	13
Guardrails	D22, D25	Week 10 (7 days)	9
Harness engineering	<i>thin — D19 schemas only</i>	D30, D31	2 + W5 project
Loop engineering	<i>mechanics — D19, D20, D24</i>	D32, D33	2 + W5 project
Context engineering	<i>scattered — D17, D18, D24, D26</i>	D36, D37	2 + W6 project

Project Ladder

Week	Project	Type
01	Model Selection Dashboard	MINI
02	Domain Fine-Tune + Quantize	PORTFOLIO
03	End-to-End Text-to-SQL	CAPSTONE
04	Production Hardening	PORTFOLIO
05	Tool-Calling Agent	PORTFOLIO
06	MCP Server + Multi-Agent Pipeline	PORTFOLIO
07	Eval & Observability Platform	PORTFOLIO
08	Scalable Agent Runtime	MINI
09	Multimodal Document Processor	PORTFOLIO
10	Red Team Your Own RAG System	PORTFOLIO
11	Deploy a Production-Ready AI Service	PORTFOLIO
12	Enterprise AI Assistant	CAPSTONE

Daily Protocol

Every teaching day runs all four steps:

1. **TEACH** the day's topic fully in chat – concept-first, plain English, FAANG examples, Architect's Lens, hands-on exercises
2. **EAGLE EYE** two-pass verification: automated keyword scan, then MANDATORY snip verification on EVERY flag. Never declare a gap without snip verification; never declare coverage without confirmation. Standard: "Don't hallucinate and don't miss anything."
3. **PDF** -> ~/Desktop/AI-ML/WeekN_DayNN_TopicName.pdf
4. **PUSH** -> week-NN/notes/ on the GitHub repository

FAANG RULE (LOCKED)

All real-world examples use ONLY Meta, Apple, Amazon, Netflix, Google, Microsoft. Hard constraint on all content.

RESUME TRIGGER

Say "Day 29 Week 5 Start" on or after 14 September 2026.

Authoritative Sources

Document	Authority over
SCHEDULE.md	All dates. If any document disagrees, this file wins.
CURRICULUM_PHASE2.md	Phase 2 content. Weeks 5–6 day-by-day; 7–12 week level.
CURRICULUM_EXTENDED.md	Weeks 10 & 11 detail.
README.md	Repository overview and progress.
Day 28 PDF §5	RESUME BRIEF — read first on 14 September.

Repository: `git@github.com:irudi2021/ai-engineering-journey.git`

Lessons: `~/Desktop/AI-ML/WeekN_DayNN_*.pdf` and `week-NN/notes/` in the repository.

Progress: 28 of 84 days complete (33%). Phase 1 sealed and verified. Phase 2 resumes Monday 14 September 2026.